Ying Feng

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Education

Massachusetts Institute of Technology — Sep 2024 – May 2028 (expected)
Ph.D. in Computer Science · Advisor: Piotr Indyk — *Cambridge, MA*

Carnegie Mellon University — Sep 2020 – May 2024
B.S. in Computer Science, with University Honors — *Pittsburgh, PA*

Research Highlights

My research develops provably efficient methods for LLM inference and retrieval.

Fast Quantization for Inference and Retrieval

- Proved that randomized Hadamard transforms preserve the error guarantees of rotation-based quantizers (e.g., EDEN, TurboQuant) while reducing rotation time from $\Theta(d^2)$ to $\Theta(d \log d)$. [4]
- Implemented Hadamard-based quantizers that nearly match the original quantizers' nearest-neighbor recall on 1536- and 3072-dim OpenAI embeddings while running 2.6–4.6× faster.

KV-Cache Compression

- Proved nearly matching upper and lower bounds on the KV-cache memory required to approximate attention during autoregressive decoding. [2]
- Showed that under the small key–query similarities measured in Llama 3.1, Qwen3, and Ministral 3, uniform token eviction is provably near-optimal, explaining why simple sampling baselines remain competitive. [13]

Vector Search and Multi-Vector Retrieval

- Built approximate nearest-neighbor indexes that answers every query correctly: a randomized construction with $d \cdot \mathrm{polylog}(n)$ search time, powered by our new fast embedding; and a deterministic construction with $d^2 + d\,\mathrm{polylog}(n)$ search time [1, 6, 14]
- Designed algorithms for multi-vector retrieval under the Chamfer similarity (ColBERT-style scoring), attaining near-optimal runtimes for underlying ℓ_p metrics, $p \in [1, 2]$. [5, 8]

Adversarially Robust Search and Sketching

- Designed vector search, regression, and frequency-estimation algorithms that stay correct under adversarial queries, using differential privacy and cryptographic techniques. [7, 9, 11]

Publications and Manuscripts

*Authors are listed alphabetically; exceptions use * to indicate first author(s).*

Conference Publications

1. **Ultra-Fast Deterministic Approximate Near Neighbor Search in High Dimensions**
   Ying Feng and Piotr Indyk
   ITCS 2027

2. **Towards Tight Bounds for Streaming Attention**
   Justin Y. Chen, **Ying Feng**, Piotr Indyk, Michael Kapralov, Ekaterina Kochetkova, and Boris Prokhorov
   SODA 2027

3. **Condition Number-Independent Sparse Linear Regression on Random Supports**
   Ying Feng
   SOSA 2027

4. **Provable Quantization with Randomized Hadamard Transform**
   Ying Feng, Piotr Indyk, Michael Kapralov, Dmitry Krachun, and Boris Prokhorov
   NeurIPS 2026

5. **Fast Approximate Lp Chamfer Distance via Lopsided Embeddings and Structured JL**
   Ying Feng and David P. Woodruff
   NeurIPS 2026

6. **Fast and Compact Random Mappings with Uniform Guarantees and Applications**
   Ying Feng and Piotr Indyk
   STOC 2026

7. **On Differential Privacy for Adaptively Solving Search Problems via Sketching**
   Shiyuan Feng, **Ying Feng**, George Z. Li, Zhao Song, David P. Woodruff, and Lichen Zhang
   ICML 2025. **Oral presentation** (top 1% of submissions).

8. **Even Faster Algorithm for the Chamfer Distance**
   Ying Feng and Piotr Indyk
   ICALP 2025

9. **Fast White-Box Adversarial Streaming Without a Random Oracle**
   Ying Feng, Aayush Jain, and David P. Woodruff
   ICML 2024

10. **A Real-Time Rescheduling Algorithm for Multi-Robot Plan Execution**
    Ying Feng*, Adittyo Paul, Zhe Chen, and Jiaoyang Li
    ICAPS 2024

11. **Improved Algorithms for White-Box Adversarial Streams**
    Ying Feng and David P. Woodruff
    ICML 2023

12. **A Fast Rescheduling Algorithm for Real-Time Multi-Robot Coordination**
    Adittyo Paul*, **Ying Feng***, and Jiaoyang Li
    SoCS 2023

Under Review

13. **Attention under Bounded Key–Query Similarity: Space Complexity and Transformer Anisotropy**
    Justin Y. Chen, **Ying Feng**, Piotr Indyk, Michael Kapralov, Ekaterina Kochetkova, and Boris Prokhorov

14. **Deterministic Approximate Nearest Neighbors in High Dimensions**
    Ying Feng and Piotr Indyk

Technical Report

- **Accelerating Scientific Research with Gemini: Case Studies and Common Techniques**
  David P. Woodruff*, et al. (including **Ying Feng**)
  arXiv preprint, 2026

Service

Conference reviewer: NeurIPS (2025, 2026), ICML (2026, **Gold Reviewer**), ICLR (2026, 2027), SODA (2026), ITCS (2026, 2027), RANDOM (2025).

Journal reviewer: SIAM Journal on Mathematics of Data Science (SIMODS).

Awards

- MathWorks Fellowship 2026 – 2027
- MIT Akamai Presidential Fellowship 2024 – 2025

Teaching

Teaching Assistant, Matrix Algebra with Applications 2023
Carnegie Mellon University *Pittsburgh, PA*

Skills

Programming: Python, C/C++.

ML & Systems: PyTorch, Hugging Face Transformers, FAISS, Slurm, Git, CUDA (basic).

Available Start and End Dates

May – September 2027.